

Smartshop v3

Software Architecture Document (SAD)

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Architecture Pattern:	Vercel Edge Serverless Microservices + Client-side PWA Shell
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1. Architectural Overview & System Topology

Smartshop v3 implements a highly decoupled, serverless-native software architecture optimized for maximum reliability and minimum network latency in mobile environments. The application utilizes a **Jamstack Client-Serverless Topology** where the frontend is compiled as a progressive React Single Page Application (SPA), served dynamically from Vercel's global CDN Edge, and communicates via asynchronous REST API calls with a unified Node.js serverless back-end gateway (`api/index.ts`).

1.1 Unified Serverless Microservices

To avoid obsolete microservice configurations (such as standard virtual machines, containerized deployments, or complex Kubernetes routing), all business logic, driver tasks, notifications, and transactions are compiled natively inside Vercel's Node.js serverless functions. This completely eliminates infrastructure overhead while guaranteeing 100% autoscale during surge order hours.

2. Full-Stack Security Architecture

Data security and transactional integrity are enforced strictly across all layers through robust encryption and verification paradigms.

2.1 Cryptographic Hash Verification

Because the app runs inside the Telegram WebApp interface, user identity is cryptographically verified upon every backend request. The server extracts the raw Telegram `initData`` query string, sorts all params alphabetically, and computes an HMAC-SHA256 hash using the private Telegram Bot Token as the secure salt. This completely prevents session spoofing or malicious request interceptions.

2.2 SQL Injection & Parameterization

Direct SQL database connections are avoided. All communications with Supabase are mediated through the official Supabase JS SDK, which uses fully parameterized PostgREST calls under the hood, neutralizing SQL injection vectors automatically.

2.3 Webhook Authenticity (Chapa Payments)

To protect against counterfeit payment confirmations, our newly established serverless Chapa Webhook listener verifies incoming callbacks through Chapa's private signature hash. Only verified requests trigger database state changes and customer notifications.

3. High-Throughput & Performance Optimization

With thousands of concurrent shoppers, the system prevents server exhaustions through three major edge caching and pooling layers.

3.1 In-Memory Sliding Window Rate Limiting

The serverless backend protects database connection quotas using a fast in-memory sliding window rate limiter keyed by client IPs. Requests exceeding 60 operations per minute are rejected instantly with an HTTP `429 Too Many Requests` code, preserving Supabase resources.

3.2 PgBouncer Connection Pooling

Serverless Node.js functions create independent database connections on the fly. To prevent connection exhaustion, Vercel connects directly to Supabase's integrated `PgBouncer port` (`6543`), which multiplexes, queues, and recycles active database connections seamlessly.

3.3 Edge CDN Caching

Static resources, scripts, and vector graphic files are aggressively cached on Vercel's Edge nodes (using strict `immutable` Cache-Control directives defined in `vercel.json`). This ensures static templates load in under `50ms` anywhere in East Africa.